

The Effects of Blended Learning Environment on the Critical Thinking and Problem-Solving Skills of
the Students

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The traditional classroom setting has long been the cornerstone of education, where students learn through a one-way dissemination of information from teachers to students. However, with the advent of technology and the increasing demand for digital literacy, the concept of blended learning has emerged as a game-changer in the education landscape. Blended learning environments, which combine traditional face-to-face instruction with online learning opportunities, have been gaining popularity in recent years due to their potential to enhance student engagement, motivation, and outcomes. One of the most significant benefits of blended learning is its impact on the development of critical skills in students. Critical skills, such as critical thinking, problem-solving, collaboration, and communication, are essential for success in today's rapidly changing and increasingly complex world. As educators seek to prepare students for the challenges of the 21st century, it is essential to understand how blended learning environments can foster these critical skills. This study will explore the effect of blended learning environments on the critical skills of students, examining the empirical evidence and theoretical frameworks that support this phenomenon. The ability to think critically is a highly valued skill in today's fast-paced and ever-changing world. As educators, it is our responsibility to equip students with the skills necessary to navigate complex information, solve problems, and make informed decisions. Blended learning environments, which combine traditional face-to-face instruction with online learning opportunities, have been touted as a potential solution to fostering critical thinking skills in students.

Research has shown that blended learning environments can provide students with greater autonomy, flexibility, and control over their learning experience, which can lead to increased engagement and motivation (Jamaluddin, M., Mustaji, M., & Bahri, B. S. 2022). Moreover, the use of online resources and multimedia tools can facilitate the development of critical thinking skills by providing students with opportunities to analyze and evaluate information, as well as construct and defend their own arguments (Alsubaie, A. H. 2020)

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In particular, studies have shown that blended learning environments can enhance critical thinking skills by promoting deeper levels of learning, such as problem-solving and decision-making (Means, M., et, al., 2013). For example, a study by Means, M.,et al. (2013) found that students who participated in a blended learning environment demonstrated significant improvements in critical thinking skills compared to those who participated in traditional face-to-face instruction.

Blended learning, thoughtfully combining the best elements of online and face-to-face education, is likely to emerge as the predominant teaching model of the future. (Yen J. C.,& Lee C. Y., 2011). Additionally, blended learning is one of the many learning models that can be applied to learning activities at this time. (Arbor S., Masitoh S., & Arianto F. 2022)

Critical thinking has drawn a lot of attention both in educational research and practice. It is observed that there are at least six ways of defining critical thinking, namely logic, skill and ability components, pedagogy, ways of being (a set of dispositions, attitudes and habits), developmental approach, and integrative approach (Kwan & Wong F., 2015). Moreover, one of the basic, constructive needs of humans, which plays a major part in their development is critical thinking. As education is one of the factors in shaping individuals' critical thinking. (Alireza, Mortezaei, Haftador., Banafsheh, Tehranineshat., Zahra, Keshtkaran., Zinat, Mohebbi., 2023).

For generations, traditional education has utilized face-to-face classes, a method in which teachers and students convene in a shared physical environment to learn together. This approach gained renewed attention during the pandemic when the global educational landscape was disrupted. Opinions about face-to-face classes vary widely, with some staunch advocates and others voicing concerns, fueling a spirited debate on the subject. In the past few years, the rise of technological innovations has resulted in an increasing dependence on digital platforms for educational use. (Aquino, T., Cabuig L.,et al, 20214). There is no doubt that utilizing recent developments in the educational process in order to achieve the education goals is one of the requirements of educational success. They bridge the gap between educational methods and technological developments. (Mohamed Al-Ghadouni., n.d.).

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As researchers continue to explore the potential of blended learning environments to foster critical thinking skills, it is essential to understand the mechanisms by which this occurs. This literature review will examine the empirical evidence and theoretical frameworks that support the impact of blended learning environments on the critical thinking skills of students.

Theme 1: Problem-Solving Skill

REFERENCE	What key concepts about your topic were highlighted? What was the research gap/ problem in literatures reviewed by the author/s?	What was/were the objectives/research questions of the study/	What was the research design of the study? Who were the participants ? What data gathering tools were used?	What were the findings/results of the study?	What was the conclusion derived? What did the study recommend ?
Abror, S., Masitoh, S., & Arianto, F. (2022). The Effect of Blended Learning on	The purpose of this problem-solving ability is to provide a way out or a solution for students in dealing with a	To determine the effect of the ability to solve problems on the subject of Islamic Cultural History. 1) How did the blended learning	The study used a quasi-experimental quantitative Approach (nonequivalent control group	The study's results based on the independent sample test showed that the pre-test scores of the two groups had no	The blended learning significantly affects problem-solving ability in the subjects History of Islamic

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Problem-Solving Ability in Islamic Cultural History Lessons. Edureligia: Jurnal Pendidikan Agama Islam, 6(01), 50-58.	problem obtained by using their thinking skills; therefore, teachers are encouraged to provide learning that leads to a thinking activity.	affects the solving abilities of students? 2) What are the impact of blended learning to the learning abilities of the students? 3) What are the benefits of blended learning to the students?	design). The data collection technique used a test (essay) given at the pre-test and post-test; the data analysis used a prerequisite test, namely, the normality test and homogeneity test, followed by an independent sample t-test. The research location was at MTsN 1 Jombang, involving 120 students, 60 students as the experimental group applying blended learning and 60 students as the control group applying	significant difference, namely accountable ($0.393 < 2.001$) with sig. (2-tailed) $0.695 > 0.05$, it can be ascertained that there is no difference between the pre-test data of the experimental and control groups. And the results of the independent sample test showed that the post-test scores of the two groups had significant differences, namely the results $t_{count} > t_{table}$ ($12.370 > 2.001$) with sig. (2-tailed) $0.000 < 0.05$. So, it can be as certain that there is a	Cultural at MTsN 1 Jombang in the experimental group compared to conventional learning in the experimental group by reviewing the post-test results of the two groups. The pre-test of the two groups was conducted to determine students' initial ability before being given treatment.
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			conventional learning.	difference between the post-test of the experimental and control groups.	
Jamaluddin, M., Mustaji, M., & Bahri, B. S. (2022). Effect of Blended Learning Models and Self-Efficacy on Mathematical Problem-Solving Ability. International Journal of Learning, Teaching and Educational Research, 21(7), 127-144.	To determine the effect of the flipped-classroom and flex blended learning models in enhancing the mathematical problem-solving ability of junior high school students. Based on this description, problem-solving is a thinking activity perceived as an effort to solve questions or issues that are considered difficult and need to be	To determine or obtain the actual value of certain objects that are not known, as well as provide appropriate conditions. 1) What are the problem-solving ability that the students used in everyday life? 2)How did the students respond to various situations? 3) What are the ability of students in enhancing their confidence?	The quasi-experimental pre-and post-test method was used to carry out this research. The sample consisted of 128 students divided into two equal groups (n1=n2=64). Self-efficacy data were collected through a questionnaire, while problem-solving ability was evaluated using validated mathematics problem test-sets.	This implies that in circumstances where students are faced with a particular problem, the higher their self-efficacy, the more positive their attitude. Conversely, it implies that the lower their self-efficacy, the higher the feeling of being threatened and the more negative the attitude. Students need to be able to overcome obstacles and not give up quickly on these online subjects. Self-efficacy is	The main objective of this research was to examine the effect of blended learning with flipped and flex models and self-efficacy, respectively, on the mathematical problem-solving abilities of Grade8 students and the interaction between these two factors. In summary, participants in the flipped class obtained higher problem-solving ability than those in the flex class.

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	resolved to achieve certain goals.			needed for students to develop their thinking processes (Chen et al., 2015; Lai & Hwang, 2016). It helps them to predict their problem-solving abilities in a controlled model as well as to detect the effects of anxiety, cognitive competence, math achievement, and self-efficacy on independent learning (Pajares & Graham, 1999).	Furthermore, those who had high self-efficacy performed better than those with low self-efficacy. There was no interaction between the learning models and self-efficacy. This indicates that its effect on the mathematical problem-solving ability does not depend on the level of self-efficacy.
Suprabha, K., & Subramonian, G. (2019). Effect of blended learning	The advantage of a problem-solving skill approach to treating children with problems is	The objective of the study was studying the effect of blended learning strategy over constructivist teaching strategy on problem solving skill of higher secondary commerce students.	The study employs pre-test and post-test non-equivalent control group design with a sample of 40 students each	School authorities start initiative to give awareness about blended learning plays important role as it is one of the modes of e-	After the experimentation of blended learning instructional strategy, the mean scores obtained for the dependent

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strategy on problem solving skill of higher secondary commerce students. i-manager's Journal on School Educational Technology, 15(2), 37-44.	that it is essentially a positive approach, which assumes that children can be taught the skill necessary to behave in different situations in an acceptable manner. Students' low-level engagement with academic activities is considering the main reason for dissatisfaction and sometimes leads to drop out of the school.	1) What are the thinking skills the student used? 2)How good is their problem-solving skills? 3)What are the student's abilities in enhancing their different skills?	in experimental group and control group. The result of the study showed that the experimental group taught through blended learning instructional strategy scored significantly more on the post-test and gain scores on problem solving skill when compared to the control group.	learning, blended learning useful for planning and analyzing the curriculum at higher secondary level and identifying the gaps where e-learning needed in the curriculum. Higher secondary students can have specific understanding about the concepts in commerce while applying blended learning strategy by which students can get the concepts more logically and explainable.	variable: problem solving skill were found to be high for both experimental and control groups. The experimental group was in more advantageous position in the mean-based approach to teaching and learning has identified the introduction of e-learning as one of the objectives of curriculum renewal. The concept of Blended learning is a pedagogical model of e-learning based on a learner centered approach to learning.
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Yeen-Ju, H. T., Mai, N., & Selvaretnam, B. (2015). Enhancing problem-solving skills in an authentic blended learning environment: A Malaysian context. International journal of information and education technology, 5(11), 841.	The development of blended learning environment through the integration of authentic learning strategies designed to enhance problem-solving skills amongst Malaysian undergraduat e students. Graduate unemployment rate in Malaysia is an increasing problem and often linked to a lack of problem-solving, critical thinking and communicatio n skills.	The objective of the subject is to expose students to the basic characteristics of multimedia elements and techniques in content creation for the web. 1) How did the students articulate their growing understanding? 2) What did the teacher's offer to support student's skills improvements? 3)How did the students improve their learning skills?	Using Coaching and Scaffolding and Integrated Authentic Assessment. The class comprised of 79 students from various ethnic backgrounds and of which 44 were female students and 35 were male students. These students had no background in multimedia and no experience using multimedia authoring tools.	This study utilized a mixed method research design whereby both quantitative and qualitative research instruments were administered to measure student attitude and perceptions towards their learning experience in the authentic blended learning environment. Data was collected via a Likert-scale survey, open-ended questionnaires and interviews with the students. The results were later triangulated to gain an in-depth insight into the students' attitudes and	The study has found that students' problem-solving skills were enhanced when the learning environment was one that incorporated authentic learning strategies and was supported by face to face as well as online learning environments. Survey results and favorable feedback from students indicate that during completing the authentic task, students were able to develop important higher order thinking skills like communication
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				perceptions of the project, working in groups, and the overall learning environment.	skills, problem-solving skills, critical thinking skills and creative thinking skills. Overall students found the learning environment to be relevant to their learning and gave them the confidence to solve similar problems in the future. This provides positive support to the use of authentic blended learning environments towards enhancing problem-solving skills among undergraduate s.
Yen, J. C., & Lee, C. Y. (2011).	To explore problem solving patterns and	The purpose of this study was to observe students' patterns of problem solving and	Quasi-experimental instruction	The descriptive statistics of the variables used in the study are	Several pedagogical implications can be drawn

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Exploring problem solving patterns and their impact on learning achievement in a blended learning environment. Computers & Education, 56(1), 138-145.	their impact on learning achievement in a blended learning environment. Students in virtual e-learning environments may experience feelings of isolation, frustration, or confusion regarding the subject matter.	their impact on learning, according to how these resources are utilized in classrooms, through mobile devices, and in web scenarios. The current research focuses on the following three questions: 1. Can any meaningful clusters of students be discovered based on the frequency of using mobile learning, web learning and classroom learning? 2. What are the problem-solving patterns based on the meaningful clusters (if any)? 3. What are the differences of learning achievement based on the meaningful clusters (if any)?	The participants of the study included 34 freshmen; 14 of whom were male and 20 females. All participants were enrolled in an "Introduction to Computer Networks" course at a science and technology institution in Northern Taiwan. Using classroom instruction, mobile and web scenarios.	shown in Table 2. They indicate that the distribution of all the variables was well within the range of the normal spread of scores, except for the frequency of learning (FOL), which peaked significantly and was somewhat skewed toward the higher end of the record.	from this study. First, 75% of the female learners were clustered in the hybrid-oriented group, while 71.4% of the male participants were in the technology-oriented group. This result complements Lee and Yuan's (in press) study in which males believed they could enjoy using technological tools, while females held more pragmatic views of technology. Blended learning should be viewed as a pedagogical approach that combines the
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					effectiveness and socialization opportunities of the classroom with the technological enhancements of the online environment
Yennita, Y., & Zukmadini, A. Y. (2021). Problem- based learning (PBL) and blended learning in improving critical thinking skills and student learning activities in biochemistry courses. In Journal of Physics: Conference	The efforts in improving critical thinking skills are carried out by applying the problem- based learning model using "blended learning" method. Evaluation of the quality of the results and the process of teaching and learning activities in biochemistry courses that have been conducted by	Improving student's critical thinking skills by requiring learning strategies that can help student to be able to develop their problem-solving skills and critical thinking skills. 1) What are the student's orientation towards the problem? 2) How did the instructors' organized students into learning? 3) How the students analyze and evaluate problem solving?	This type of research is classroom action research. This research will be planned in two cycles. Each cycle consists of four stages, namely planning, implementatio n, observation, and reflection. The instrument used to collect data was a critical thinking test sheet and an observation sheet. This activity	PBL learning that is implemented in biochemistry courses, is not only done conventionally, but also using modern technology. The use of technology makes student learning activities increase, thus providing an increase in students' thinking skills. This is in accordance with the opinion which states that modern technologies are assimilated with	It can be concluded that problem-based learning that is applied by using the blended learning method can improve student learning activities, lecturer teaching activities, as well as critical thinking skills of students. The application of problem-based learning (PBL) using the blended learning

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Series (Vol. 1731, No. 1, p. 012007). IOP Publishing.	the instructor shows that students' critical thinking skills are still very low. This can be seen from the lack of student participation in asking students questions, answering questions, responding to information, and not being able to use their thinking skills to solve problems. Students have not been able to express arguments and connect one concept with another concept based on their analysis and critical thinking about an		involved 30 students taking biochemistry courses in the Program Studi S-1 Pendidikan Biologi, Universitas Bengkulu, Indonesia.	the instructional design models for students to construct their knowledge, meaningful learning, and critical thinking.	method can improve critical thinking skills and student learning activities in biochemistry courses.
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	existing problem.				
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Theme 2: Critical Thinking Skill

REFERENCE	What key concepts about your topic were highlighted? What was the research gap/problem in literatures reviewed by the author/s?	What was/were the objectives/research questions of the study/	What was the research design of the study? Who were the participants? What data gathering tools were used?	What were the findings/results of the study?	What was the conclusion derived? What did the study recommend?
Abdul, Haris, Sunubi., Bachtiar, Bachtiar. (2022). Blended Learning Method in Enhancing Students'	Blended Learning Method, Critical Thinking Skills,	The purpose of this study is to ascertain the potential and difficulties associated with the blended learning	This type of research uses a qualitative case study methodology	The study reveals that blended learning in junior high schools in	The study conclude that students' critical thinking abilities can

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Critical Thinking Skills: Challenges and Opportunities. Al-Ishlah, 14(4):6817-6824. doi: 10.35445/alishlah.v14i4.2163	Challenges and Opportunities Growing technological sophistication presents both opportunities and problems for educational institutions to implement technology-based learning materials, enabling learning activities to take place anytime, anywhere, without being limited by time, place, or number. Though,	approach in raising students' critical thinking abilities in South Sulawesi Junior High Schools.	y. With the use of data collection techniques using observation, interviews, and documentation. In 2022, two principals, two English teachers, and forty-five students participated in an English-focused study at South Sulawesi's Senior High School. Methods of gathering data include observation to assess the state of the school and its infrastructure, interviews	Pinrang has strengths and weaknesses. Teachers play a crucial role in enhancing the learning process, listening to students' opinions, and promoting communication. Blended learning aims to overcome the weaknesses of face-to-face learning by creating interesting material and fostering critical thinking. To succeed, student learning activities should be varied, not monotonous, and use	be enhanced by the blended learning approach. To increase student interest in the content offered in Google Classroom and increase the likelihood that they will assist in solving difficulties, teachers must be able to devise fresh and innovative teaching methods. Certain concerns, such a lack of self-awareness, a lack of technological literacy and competency,
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	many educators continue to overlook the benefits of technology-based learning resources.		with instructors and students to gather information about blended learning, and documentation of the test results for certain subjects.	technology in a balanced way. Teachers must also develop an atmosphere of communication, respect, and equality.	network challenges, and a lack of information and communication technology tools for some people, need to be resolved. When it is believed that in-person instruction is less successful, blended learning can also be utilized to reach a larger number of pupils.
Ahmad, Fauzan., Rispawati, Rispawati., Lalu, Sumardi. (2023). Blended Learning Models Using Moodle	Blended Learning, Moodle SPADA UNRAM, and Critical Thinking	Aims to increase knowledge of blended learning applications. models utilizing moodle Spada Unram to	This study makes use of a quantitative methodology for the study for 60	The findings of the study are 60 students enhance students' understanding of college-	The content learning system Moodle Spada Unram has features such as

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SPADA Unram to Enhance Student Critical Thinking PPKn Study Program FKIP University of Mataram. Jurnal ilmiah profesi pendidikan, 8(1b):482-491. doi: 10.29303/jipp.v8i1b.1229	Overcoming challenges in the classroom like restrictions on the classroom, level presence of the lecturer, and level presence of the students. Blended learning, Moodle, SPADA, UNRAM, and critical thinking in the classroom. Thoughts are explained permit the growth of dissemination of information, dialogue,	encourage pupils' critical thinking enrolled in courses at the FKIP and PPKn universities in Mataram in order to supply the most productive and successful educational experience feasible by utilizing a variety of teaching techniques.	students in Unram. The kind of study that was conducted in This study uses a quasi-experimental design. There was no need for a collective control because this was a group experiment with only one group participating in the investigation. process.	level theory and improve their analytical reasoning abilities. There critical thinking adopts an alternative perspective, evaluating arguments, drawing conclusions using inductive or judging reasoning, using deductive reasoning, and deciding on solutions.	interactive media, competency basics, description eye lessons, topic theory lessons, way learning, and assessment. This approach uses a computer to deliver college-level theory in video format to improve students' critical-thinking abilities. Analytical, reasoning, questioning, probing, and other abilities are all necessary for critical thinking. The skills part of critical
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	and involvement of students later.				thinking involves analyzing arguments, drawing conclusions through inductive or deductive reasoning, evaluating, or judging, and coming to decisions or solving issues.
Alireza, Mortezaei, Haftador., Banafsheh, Tehranineshat., Zahra, Keshtkaran., Zinat, Mohebbi. (2023). A study of the effects of blended learning on university students' critical thinking: A systematic review. Journal of education and health promotion, 12(1):95-95. doi:	Critical thinking; distance learning; online learning; systematic review. The goal of education has shifted from being the mere transfer of knowledge at the expense of	Dynamic teaching approaches foster students' capacity to explore and recognize their own learning requirements, to engage in critical thinking, to actively participate in their education, to be more proactive in many contexts, and to enhance their problem-solving abilities.	The QUOROM statement checklist, an evidence-based framework that regulates reporting in systematic reviews and meta-analyses, served as the guide for conducting this	Flipped classroom learning, a subcategory of blended learning, combined the best aspects of electronic and traditional learning. The results of these investigations demonstrated that flipped classroom instruction	The conclusion showed that, by combining the two methods of lecturing and electronic learning, blended learning and its subcategory of flipped classroom enable teachers to use the

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10.4103/jehp.jehp_665_22	developing intelligent and creative individuals to being the transformation of students from passive learners into active learners who seek knowledge effects of the flipped classroom model (one of the subcategories of blended learning) on students' critical thinking and problem-solving skills.		systematic review.	improved students' critical thinking across majors and, as a result, could be implemented in academic settings.	advantages of both approaches and encourage student-centre learning. This causes blended learning to be a more effective method for improving students' critical thinking, in terms of both disposition and skills. It suggested that future studies compare the effectiveness of blended learning with other modern methods of learning.
Alsubaie, A. H. (2020). The effect of blended	Blended learning, critical	The process of acquiring thinking abilities, especially	The researcher used a semi-	Indicates that there is no statistically	This study looks at how critical

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learning in developing critical thinking skills among 8th grade students in history subject. <i>Journal of Positive Psychology and Wellbeing</i> , 3 (4), 70-78.	thinking skills, Eighth grade, History Subject The concept of education encompass es not just electronic technology but also a complete redesign of the process, reorganizati on of teacher-student roles, and a new educational framework. Traditional education remains the most impactful, and its integration faces challenges	critical thinking abilities, which are necessary to do analysis and assessment of current knowledge and to produce new, useful knowledge. The study's main challenge is to teach eighth-grade students how to integrate technology into their education by using integrated education to help them develop critical thinking abilities considering the various obstacles that the educational process faces.	experimental design to investigate the impact of integrated education on critical thinking in students, ensuring complete data and effective answers to the study's objectives.	significant difference between the post-performance of the critical thinking skills test and the delayed test. This can be attributed to the integrated education strategy's effectiveness in preserving students' critical thinking abilities and in giving them knowledge, facts, and ideas; its role in preserving and sequentially organizing them within the cognitive structure; and its installation method, which is in line with the aspects of the	thinking centers are set up in Jordanian university curricula. The type of the activities, extra readings, the arrangement of the instructional materials, instructor motivation, and the teaching approach for integrated education are some of the aspects that may be responsible for the disparity. According to the report, modern education seeks to capitalize on factors such
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	due to differing mental capacities.			human mind that give education meaning. The emphasis on meaning has made it clear how important it is for pupils to acquire critical thinking techniques and to grow as individuals.	as individual student variances, absence of in-person instruction owing to the Corona pandemic, and distance learning technologies. The report recommends instituting integrated education, allocating a fair amount of time to each subject area, providing teacher preparation programs, and assessing the critical thinking abilities of Jordanian university students.
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Kwan, & Wong, F. (2015). Effects of the constructivist learning environment on students' critical thinking ability: Cognitive and motivational variables as mediators.	A constructivist learning effect on critical thinking ability and cognitive strategies mediated the relationship in critical thinking. The constructivist learning environment excluded blended learning to its effects to critical thinking.	1. Is cognitive strategies significantly mediate relationship between following: a. constructivist learning environment b. critical thinking ability 2. Is goal orientation significantly mediate relationship between following: a. constructivist learning environment b. critical thinking ability 3. What are the variables correlation in prediction in the following: a. constructivist learning environment. b. critical thinking ability c. goal orientations	Descriptive and bivariate correlations. A sample consisted of 967 grade nine students who were studying Liberal Studies or Integrated Humanities from seven secondary schools in Hong Kong. A questionnaire made up of items adopted from the Constructivist Learning Environment Survey	Each variable has positive correlation which means they that one variable positively predicts the other variable and the cognitive strategies and goal orientation mediate relationship between constructivist learning environment and critical thinking ability.	The direct effect of constructivist learning environment on critical thinking ability and its indirect effect on critical thinking ability. The results of this study have provided some information supporting the following issues related to developing critical thinking ability.
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		d. cognitive strategies			
Lin, F., & Huang, H. M. (2024). A blended learning intervention involving Problem-Based and Team-Based learning to enhance students'. . . ResearchGate. https://doi.org/10.9790/7388-1401013442	The use of blended learning to enhance the critical thinking of a student. An inadequate learning motives and a reluctance to participate in active thinking and learning. To adapt the teaching approach, problem-based learning (PBL) was included pupils must be split out into several discussion-	To investigate the effects of a blended learning intervention involving problem-based (PBL) and team-based (TBL) learning on students' critical thinking skills, disposition, and learning outcomes. Here are the research questions of the study: (1) How does a blended PBL-TBL learning intervention in child health nursing influence two-year nursing students' critical thinking skills? (2) How does a blended PBL-TBL learning intervention in child health nursing influence two-year nursing	A pre-posttest, single-group, quasi-experimental design was used. The individuals taking part were chosen included two-year nursing students enrolled in a two-credit Child Health course, selected using convenience sampling. A Northern Taiwanese university's nursing program (N = 51).	Students' total CTT-II and CTDS scores increased considerably because of the intervention; the "deduction" dimension showed the greatest improvement. The pre-posttest changes were statistically significant for all four variables, indicating that the intervention also considerably enhanced learning outcomes.	The study highlights the benefits of collaborative TBL and PBL in large class settings, promoting idea integration, deepening education, and enhancing problem-solving skills. It emphasizes the importance of speaking, listening, and expressing in TBL over passive learning. However, the findings cannot be applied to other fields or classroom

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	focused small groups.	students' critical thinking disposition? (3) How does a blended PBL-TBL learning intervention in child health nursing influence two-year nursing students' learning outcomes?			contexts and can only serve as an empirical foundation for blended PBL-TBL learning interventions.
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Theme 3: Blended Learning

REFERENCE	What key concepts about your topic were highlighted? What was the research gap/ problem	What was/were the objectives/research questions of the study/	What was the research design of the study? Who were the participants? What data gathering	What were the findings/results of the study?	What was the conclusion derived? What did the study recommend?
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	in literature s reviewe d by the author/s ?		tools were used?		
Aquino, T., Cabig, L., Guevarra, D. T., Marcaida, J. L. M., Montemayor, J. N., Pamintuan, J. A., Romero, N. E. O., Santos, J. M., Tadeo, A., & Villamor, A. L. Jr. (2024). Measuring the impact of face- to-face learning on HUMSS students' school productivity. Ignatian International Journal for Multidisciplinary Research	In Blended learning face to face is integrated to online learning,in this study the authors objective is to measure the productivit y of humss students.Th e gap of the study is it does not research further the effect of online learning.	This research explores the impact of face- to-face learning on the productivity of Grade 12 Humanities and Social Sciences Strand students, concentrating on multifaceted aspects such as academic performance, completion of performance tasks, overall academic outcomes, student engagement, extra-curricular activities, and	Through a quantitativ e methodolo gy employing random sampling (n=54) via an online survey, the study systematic ally investigates the productivit y of Grade 12 Humanities and Social Sciences students. The researchers	Weighted means ranging from 3.43 to 3.49 and an overarching weighted mean of 3.45, indicative of a unanimous 'strongly agree' sentiment. These findings underscore the affirmative influence of physical classroom presence on various dimensions	The subscale "Effect of face-to- face learning on Students Grades" yielded a weighted mean of 3.49 with a standard deviation of 0.58, indicating a moderate to high level of agreement among the respondents . This suggests a positive

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		school attendance rates. This study aims to measure the impact of face-to-face learning on students' productivity in school. The study specifically aims to give answers to the following questions: 1. What is the effect of face-to-face learning in students' grades? 2. What impact does face-to-face learning have on students' ability to complete performance tasks? 3. How does physical presence in a classroom setting influence overall academic outcomes? 4. What is the level of student engagement and	harnessed descriptive statistics for data analysis, unveiling a robust consensus among the participant s. The respondent s overwhelmingly expressed a positive outlook on the impact of face-to-face learning.	of student experience. The study asserts that the tangible presence in a classroom setting significantly contributes to the academic success, heightened engagement , and overall productivity of Grade 12 Humanities and Social Sciences Strand students.	perception among the surveyed individuals regarding the efficacy of face-to-face learning in influencing academic performance. Expand the demographic scope of the study by including a more diverse sample in terms of geographical location, socio-economic background, and cultural factors. This would contribute to a more
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		participation in extracurricular activities in face-to-face learning? 5. What is the effect of face-to-face learning in students' school attendance rates?			comprehensive understanding of how various student populations perceive and benefit from face-to-face learning.
Jones. (n.d.). The Implications of Blended Learning in Today's Classroom: A Look into the History, Views, Impacts, and Research. nwcommons.nwciowa.edu.	The Impact of Impaction of Blended learning, however the study did not question about the effect of blended learning in critical thinking	This review of literature aims to explore this transformation by looking into the history leading up to the regular use of technology in the classroom, as well as the theoretical frameworks that support the premise of blended learning.	The design of the study is a descriptive-qualitative design, the study conducted students, teachers, alumni in northern west college since it is qualitative. The data gathering tool used is by implementation of	Blended learning offers a myriad of positive outcomes when implemented well, there are several opposing viewpoints addressed throughout the literature. One of the most common concerns addressed in the literature is the	Blended learning has the potential to completely transform both teaching and learning. While this review of literature seems to have merely scratched the surface, it is clear that not only do abundance

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			proposed implication of blended learning integrating the blooms taxonomy concept.	perception that by incorporating online components in the classroom, learning will become a much more impersonal activity.	theoretical frameworks support the premise of blended learning, but also that several models have been developed which encompass some of the most effective elements of blended learning It is important to remember, however, that despite these criticisms, blended learning can have tremendous impacts on students, teachers,
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					and districts. Included in this list of impacts are increased opportunities for collaboration, communication, and community; higher levels of student engagement; positive shifts in student and teacher roles; more opportunity for differentiation and personalization; and district level considerations, thus it needs for further research.
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Lu, D. (2021, June 25). Students' Perceptions of a Blended Learning Environment to Promote Critical Thinking. <i>Frontiers in Psychology</i>. https://doi.org/10.3389/fpsyg.2021.696845	This study has some implications for critical thinking cultivation in the instruction of specific disciplines relating in our study. The study gaps our study in perception of students in critical thinking, and the environment where we will conduct the topic.	The purpose of the study was to find out students' perceptions of the learning environment concerning its design and its influence on their critical thinking. By adopting the mixed method, this study aims to answer the following two questions: 1. What are students' perceptions of the blended learning environment to promote critical thinking? 2. How do students perceive the impact of the blended learning environment on the development of their critical thinking	Both quantitative and qualitative data were collected for this study. In terms of quantitative analysis, descriptive statistics were used to describe the means, standard deviations. As for qualitative data, the recordings of the interviews were transcribed for content analysis. The participant s of the study involved a total of 90	-The overall mean score was 3.86 (SD = 0.79), suggesting that students were generally satisfied with the pedagogical design. Item 1 (M = 3.98, SD = 0.80) (The learning objectives are clearly stated), Item 4 (M = 3.93, SD = 0.83) (Expectation s of assignments are clearly stated), and Item 5 (M = 4, SD = 1.00) (Activities are planned carefully) got particularly high scores, which indicates	the data from the interview suggest that students could perceive their critical thinking development in different thinking dimensions. Furthermore, according to the students' opinion, their development in critical thinking was also manifested in their writing and even transferred to other activities.
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			non-English major students (33 males and 57 females) aging from 18 to 20 in 2020. The questionnaire is adapted from the Web-Based Learning Environment Instrument (WEBLEI) was used to elicit the information of students' perception of the learning environment.	that students were aware of the careful design of the activities, content, and context. In sum, students' ratings on different dimensions of the questionnaire suggest that students perceived the productiveness of the learning environment in a generally positive way. This result is consistent with the studies exploring students' perceptions of the	The result of the study suggests that students are generally satisfied with the design of the learning environment, and students considered the learning environment helpful in improving critical thinking. Even though the study contributed to the instructional design aiming at critical thinking promotion in a blended learning
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				blending learning environment in general.	environment, some limitations should be noted.
Mariamoutaoukil. (2022). Impact of blended learning. In Technology and the Curriculum: Summer 2022.	One such use of technology, online classrooms, are potentially powerful environments where innovation practices and new relationships can be created that can have a significant impact on learning.	This paper aims to explore how blended learning has impacted students by looking into its history, some theoretical frameworks that supports blended learning.	They began the implementation of their model by creating learners' profiles for each student based on information obtained via survey, observation, on students learning styles, preferences, attitudes toward learning expectation.	The use of new technologies in blended learning allows students to gain experiences based on their preferences, enhance their life skills, and develop emotionally and physically.	By combining classroom and online learning, students not only have additional opportunities to interact with each other and with teachers but also gain access to other people through online platform. There is still further research that needs to explore various other topics

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					such as its impact on ethnic demographics, financial impacts on districts and impact of the flexibility of blended learning on schools' schedules, calendars, and classrooms.
Mishra, S, & Khozael, S. O. Blended learning and critical thinking in higher education: A review of the literature. Journal of Interactive Learning Research, 27(3), 171-185	With technology innovation over the year's reliance on the traditional face-to-face learning is being shifted knowledge delivery and fostering	This paper aims to examine and review of previous studies carried out based on the perceptions of students in blended learning environments and its effectiveness in enhancing the learning process.	Desk research technique was conducted for the study which basically involving data form the existing online library which covered several academic data bases	There was no difference in learning outcomes between groups. The CMPBL group spent significantly more time on learning than the TPBL group.	The richness of the vast literature available has illustrated several implications at the end of this review. Higher educational institutions must look beyond the traditional

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	an innovative learning environment.		such as ScienceDirect, Academia, and Wiley. It has a participant of computer assisted, computer management system, education. Higher education, students, tertiary education and undergraduate. The research was then repeated. Adding the other key words such as case study, perceptions and		boundaries of classroom instruction and engage in augmenting their current best practices with new advances in learning and collaboration
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			learning methods.		
Mohamed Al-Ghadouni. (n.d.). Critical thinking and Blended E-learning: A Review of Scientific Studies. Revistaargintinadeclinicapsicol ogica.com	The concepts of E- mastering are innovation that is ordered in a method of excellent ability from an instructive point of view, and it is one of the graces of the essential examinati on of the learning generation in several years maximum.	Exploring the subject, capturing essential speculation and investigating modalities.	For the method prism convention is being followed where the students are being provided by writing exams diagnosed with three fundament al hubs: underline, online instructors, and intuitive instructiona l programs gets knowledge about conditions.	The propose taken: the dominant subjects and exams sub- subjects, most e- studying methodologi es were investigated, most hypothetical structures related to e- know, and the typology of exploration.	E-learning- benefiting from some of its application but we need more deeper research within the traditional classroom, under the supervision and guidance of the teacher.

Research Gap

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While there is a growing interest in blended learning, there is a need for more research on how to scale and sustain effective blended learning environments that promote critical thinking skills. Current measures of critical thinking skills may not capture the complexity and depth of critical thinking. This is helpful for more research on developing nuanced measures of critical thinking skills that can be used to assess student learning outcomes in blended learning environments. With the lack of knowledge and tools, we will not be able to scale and sustain effective blended learning environments that promote critical thinking skills, that identify the key factors that influence the success of these initiatives. And developing more nuanced measures of critical thinking skills cannot be met in this research.

References

- Abror, S., Masitoh, S., & Arianto, F. (2022). The Effect of Blended Learning on Problem-Solving Ability in Islamic Cultural History Lessons. *Edureligia: Jurnal Pendidikan Agama Islam*, 6(01), 50-58.
- Abeysekera, I., & Ranasinghe, T. G. (2017). Blended learning approach to enhance critical thinking skills of undergraduate students. *International Journal of Education and Development using Information and Communication Technology*, 13(2), 181-194.
- Adedoyin, O. A., & Idowu, O. M. (2022). The effects of blended learning on students' critical thinking skills in social studies in Nigerian secondary schools. *Journal of Educational Technology Development and Exchange (JETDE)*, 15(2), 127-142.
- Adedoyin, O. A., & Antonye, S. A. (2021). The effects of blended learning on students' critical thinking skills in social studies. *Journal of Educational Technology Development and Exchange (JETDE)*, 14(2), 147-162.
- Ahmad, Fauzan., Rispawati, Rispawati., Lalu, Sumardi. (2023). Blended Learning Models Using Moodle SPADA Unram to Enhance Student Critical Thinking PPKn Study Program FKIP University of Mataram. *Jurnal ilmiah profesi pendidikan*, 8(1b):482-491. doi: 10.29303/jipp.v8i1b.1229
- Alireza, Mortezaei, Haftador., Banafsheh, Tehranineshat., Zahra, Keshtkaran., Zinat, Mohebbi. (2023). A study of the effects of blended learning on university students' critical thinking: A systematic review. *Journal of education and health promotion*, 12(1):95-95. doi: 10.4103/jehp.jehp_665_22

EFFECTS OF BLENDED LEARNING ENVIRONMENT

- Alsubaie, A. H. (2020). The effect of blended learning in developing critical thinking skills among 8th grade students in history subject. *Journal of Positive Psychology and Wellbeing*, 3(4), 70-78.
- Al-Bataineh, A. T., & Al-Qudah, J. S. (2017). The blended learning approach and its effects on the development of critical thinking skills among university students: A case study in Jordan. *International Journal of Instruction*, 10(2), 117-132.
- Al-Balawi, A. M., & Al-Sobki, A. H. (2021). The impact of blended learning on critical thinking skills among Saudi university students. *International Journal of Instruction*, 14(3), 237-254.
- Al-Ghadouni, M. (2020). Critical Thinking and Blended E-learning: A Review of Scientific Studies. *Revista Argentina de Clínica Psicológica*.
- Al-Sadiq, A. A., & Al-Emran, A. M. (2018). The impact of blended learning on the development of critical thinking skills of EFL learners. *The Journal of Educational Thought*, 7(2), 222-240.
- Al-Sunbul, A., & Morris, S. (2016). Blended learning and the development of critical thinking skills in EFL learners. *International Journal of Language Studies*, 10(4), 75-90.
- Akyüz, H. İ., & Samsa, S. (2009). The effects of blended learning environment on the critical thinking skills of students. *Procedia: Social & Behavioral Sciences*, 1(1), 1744–1748.
<https://doi.org/10.1016/j.sbspro.2009.01.308>
- Aquino, T., Cabig, L., Guevarra, D. T., Marcaida, J. L. M., Montemayor, J. N., Pamintuan, J. A., Romero, N. E. O., Santos, J. M., Tadeo, A., & Villamor, A. L. Jr. (2024). Measuring the impact of face-to-face learning on HUMSS students' school productivity. *Ignatian International Journal for Multidisciplinary Research*
- Arwaty, N., Mustamin, K., & Rahmah, A. (2023). Effectiveness of online blended learning and critical thinking skills of biology students in Indonesia: Meta-Analysis study. *Jurnal Penelitian Pendidikan IPA*, 9(12), 1284–1291. <https://doi.org/10.29303/jppipa.v9i12.5761>
- B, Hajrezayi, H, Roshani, alibinasi, M, Shahalizade, M, Zeynali, Mehdi, Badali. (2015). Effectiveness of blended learning on critical thinking skills of nursing students. *Journal of Nursing Education*, 4(1):49-59.
- Chen, T.-W. (2018). The effects of blended learning on the development of students' critical thinking skills in a teacher education program. *Journal of Computers in Education*, 7(2), 141-158.
- Chen, Y., & Liu, X. (2020). Effects of blended learning with flipped classroom on students' critical thinking skills and self-efficacy in an EFL context. *System*, 92, 102242.
- Cho, Y., & Cho, H. (2016). The effects of blended learning on critical thinking skills and student engagement in a statistics course. *The Journal of Educational Research*, 109(4), 425-439.
- Chiu, Y. H., & Shih, Y. L. (2012). The effects of blended learning with problem-based learning on students' problem-solving skills and self-efficacy. *Journal of Educational Technology Development and Exchange (JETDE)*, 5(3), 187-200.

EFFECTS OF BLENDED LEARNING ENVIRONMENT

- Dan, Lu. (2021). Students' Perceptions of a Blended Learning Environment to Promote Critical Thinking. *Frontiers in Psychology*, 12:696845-696845. doi: 10.3389/FPSYG.2021.696845
- Desi, Kusumawati., Purwandari, Purwandari., Titin, Kuntum, Mandalawati. (2023). Development of Blended Learning Modules to Improve Critically and Creativity Thinking of Students. *Al-Ishlah*, 15(2):2425-2432. doi: 10.35445/alishlah.v15i2.2737
- Eva, Y., W., Wong., Theresa, Kwong., Chi-Fai, Chan. (2021). Blended Learning Approach to Enhance Reflective Higher Order Cognitive Thinking Skills in Students. 121-132. doi: 10.1007/978-3-030-72216-6_8
- Eline, Yanty, Putri, Nasution., Laras, Sintia., Rahmi, Anggia, Putri. (2022). The Effect of Blended Learning Models Assisted by Video Tutorials on Students' Critical Thinking Ability in Mathematics Learning. *Logaritma*, 10(01):1-22. doi: 10.24952/logaritma.v10i01.5186
- Ferguson, R., & Owston, R. (2017). Blended learning: Student engagement and outcomes in a problem-based learning unit. *Higher Education Research & Development*, 36(3), 545-558.
- Green, K. C. (2018). Blended learning design for critical thinking in higher education. *TechTrends*, 62(4), 437-444.
- Haghparast, Nasaruddin, & Abdullah. (2014). Cultivating critical thinking through e-learning environment and tools: A review. *Scholar.google.com*.
- Hasanah, Hasanah., Muh, Nasir, Malik. (2020). Blended learning in improving students' critical thinking and communication skills at University. *Cypriot Journal of Educational Sciences*, 15(5):1295-1306. doi: 10.18844/CJES.V15I5.5168
- Hwang, G., & Lai, H.-Y. (2015). The effectiveness of blended learning on students' critical thinking skills in a science classroom. *Computers & Education*, 88, 121-134.
- Hwang, W., & Choi, J. (2019). The effects of a blended learning approach with problem-based learning on students' problem-solving skills and critical thinking dispositions. *Journal of Educational Technology Development and Exchange (JETDE)*, 12(2), 239-254.
- Im, Y., & Lim, S. (2019). The effects of blended learning on students' critical thinking and problem-solving skills in a science classroom. *Journal of Educational Technology Development and Exchange (JETDE)*, 12(1), 181-194.
- Jamaluddin, M., Mustaji, M., & Bahri, B. S. (2022). Effect of Blended Learning Models and Self-Efficacy on Mathematical Problem-Solving Ability. *International Journal of Learning, Teaching and Educational Research*, 21(7), 127-144.
- Jang, H. J., & Cho, H. J. (2017). Blended learning and its effects on critical thinking skills of nursing students. *Journal of Nursing Education*, 56(10), 538-544.
- Johnson, R. B., & Smith, K. A. (2015). The role of blended learning in a problem-based learning curriculum. *Interdisciplinary Journal of Problem-Based Learning*, 9(2), 3-24
- Jones. (2019). The Implications of Blended Learning in Today's Classroom: A Look into the History, Views, Impacts, and Research. *nwcommons.nwciowa.edu*

EFFECTS OF BLENDED LEARNING ENVIRONMENT

- Khaled, N., Alotaibi. (2013). The Effect of Blended Learning on Developing Critical Thinking Skills. 2(4):176-. doi: 10.11648/J.EDU.20130204.21
- Khalil, M. I., & Mohamad, N. H. (2013). Blended learning: A review of literature on the effectiveness of blended learning in secondary education. *International Journal of Instruction*, 6(1), 95-115.
- Kwan, & Wong, F. (2015). Effects of the constructivist learning environment on students' critical thinking ability: Cognitive and motivational variables as mediators. Scholar.google.com.
- Lai, C. L., & Hwang, G. J. (2016). A blended learning approach to enhance critical thinking skills and problem-solving abilities in atmospheric science education. *Journal of Geoscience Education*, 64(1), 42-52.
- Lin, C. H., & Yeh, S. C. (2014). The effects of blended learning on EFL learners' critical thinking skills and problem-solving abilities. *International Journal of Emerging Technologies in Learning (IJETL)*, 9(4), 23-34.
- Lin, F., & Huang, H. M. (2024). A blended learning intervention involving Problem-Based and Team-Based learning to enhance students. ResearchGate
- Liu, X., & Li, Y. (2014). A blended learning approach to develop critical thinking and problem-solving skills of EFL learners. *Theory and Practice in Language Studies*, 4(2), 437-444.
- Lu, D. (2021). Students' Perceptions of a Blended Learning Environment to Promote Critical Thinking. *Frontiers in Psychology*. <https://doi.org/10.3389/fpsyg.2021.696845>
- Mariammoutaoukil. (2022). Impact of blended learning. In *Technology and the Curriculum: Summer 2022*. Pressbooks.
- Means, B., Toyama, Y., Murphy, R.F. and Baki, M. (2013) The Effectiveness of Online and Blended Learning A Meta-Analysis of the Empirical Literature. *Teachers College Record*, 115, 1-47.
- Mishra, S., & Khobre, A. (2014). Blended learning: An effective approach for developing critical thinking skills in technical education. *International Journal of Innovative Research and Development*, 3(10), 248-252.
- Mishra, S., & Khozaei, S. O. (2016). Blended learning and critical thinking in higher education: A review of the literature. *Journal of Interactive Learning Research*, 27(3), 171-185.
- Morris, J. A., Belz, M. A., Woods, D. R., & Reigeluth, C. M. (2005). Blended learning: Research perspectives. *Journal of Online Learning Research*, 1(1), 5-25.
- Mutua, M. W., & Wanjiru, E. W. (2017). The impact of blended learning on critical thinking skills of pre-service teachers in Kenya. *International Journal of Instruction*, 10(3), 189-202.
- Sunubi, A. H., & Bachtiar, B. (2022). Blended Learning Method in Enhancing Students' Critical Thinking Skills: Challenges and Opportunities. *Al-Ishlah : Jurnal Pendidikan*, 14(4), 6817-6824. <https://doi.org/10.35445/alishlah.v14i4.2163>
- Yen, J. C., & Lee, C. Y. (2011). Exploring problem solving patterns and their impact on learning achievement in a blended learning environment. *Computers & Education*, 56(1), 138-145.

